> Link Kaggle: https://www.kaggle.com/amitabhajoy/bengaluru-house-price-data

Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load dataset

In [2]:
```
path = "Bangalore_House_Price_data/Bengaluru_House_Data.csv"
df_raw = pd.read_csv(path)
df_raw.shape
```

Out[2]:
```
(13320, 9)
```

In [3]:
```
df_raw.head()
```

Out[3]:

	area_type	availability	location	size	society	total_sqft	bath	balcony
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	Coomee	1056	2.0	1.0
1	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	Theanmp	2600	5.0	3.0
2	Built-up Area	Ready To Move	Uttarahalli	3 BHK	NaN	1440	2.0	3.0
3	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	Soiewre	1521	3.0	1.0
4	Super built-up Area	Ready To Move	Kothanur	2 BHK	NaN	1200	2.0	1.0

In [4]:
```
df_raw.tail()
```

Out[4]:

	area_type	availability	location	size	society	total_sqft	bath	balcor
13315	Built-up Area	Ready To Move	Whitefield	5 Bedroom	ArsiaEx	3453	4.0	0
13316	Super built-up Area	Ready To Move	Richards Town	4 BHK	NaN	3600	5.0	Na
13317	Built-up Area	Ready To Move	Raja Rajeshwari Nagar	2 BHK	Mahla T	1141	2.0	1
13318	Super built-up Area	18-Jun	Padmanabhanagar	4 BHK	SollyCl	4689	4.0	1
13319	Super built-up Area	Ready To Move	Doddathoguru	1 BHK	NaN	550	1.0	1

Exploratory Data Analysis (EDA)

In [5]:
```
df = df_raw.copy()
```

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13320 entries, 0 to 13319
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   area_type     13320 non-null  object 
 1   availability  13320 non-null  object 
 2   location      13319 non-null  object 
 3   size          13304 non-null  object 
 4   society       7818 non-null   object 
 5   total_sqft    13320 non-null  object 
 6   bath          13247 non-null  float64
 7   balcony       12711 non-null  float64
 8   price         13320 non-null  float64
dtypes: float64(3), object(6)
memory usage: 936.7+ KB
```

In [8]:
```
df.describe()
```

Out[8]:

	bath	balcony	price
count	13247.000000	12711.000000	13320.000000
mean	2.692610	1.584376	112.565627
std	1.341458	0.817263	148.971674
min	1.000000	0.000000	8.000000
25%	2.000000	1.000000	50.000000
50%	2.000000	2.000000	72.000000
75%	3.000000	2.000000	120.000000
max	40.000000	3.000000	3600.000000

In [9]:

```
def value_count(df):
  for var in df.columns:
    print(df[var].value_counts())
    print("--------------------------------")

value_count(df)
```

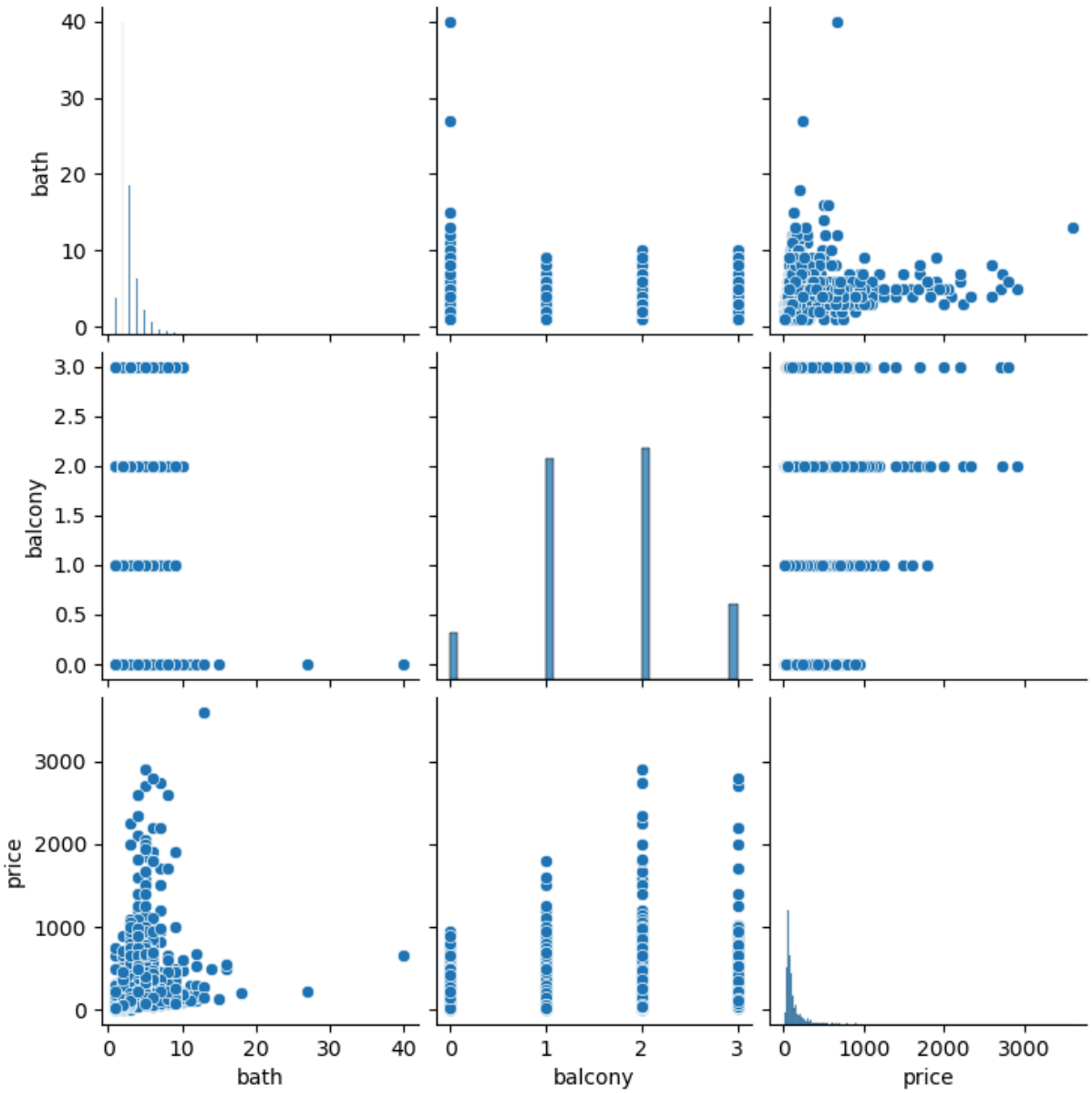
```
area_type
Super built-up  Area    8790
Built-up  Area          2418
Plot  Area              2025
Carpet  Area              87
Name: count, dtype: int64
--------------------------------
availability
Ready To Move    10581
18-Dec             307
18-May             295
18-Apr             271
18-Aug             200
                 ...
15-Aug               1
17-Jan               1
16-Nov               1
16-Jan               1
14-Jul               1
Name: count, Length: 81, dtype: int64
--------------------------------
location
Whitefield                        540
Sarjapur  Road                    399
Electronic City                   302
Kanakpura Road                    273
Thanisandra                       234
                                 ...
Bapuji Layout                       1
1st Stage Radha Krishna Layout      1
BEML Layout 5th stage               1
singapura paradise                  1
Abshot Layout                       1
Name: count, Length: 1305, dtype: int64
--------------------------------
size
2 BHK         5199
3 BHK         4310
4 Bedroom      826
4 BHK          591
3 Bedroom      547
1 BHK          538
2 Bedroom      329
5 Bedroom      297
6 Bedroom      191
1 Bedroom      105
8 Bedroom       84
7 Bedroom       83
5 BHK           59
9 Bedroom       46
6 BHK           30
7 BHK           17
1 RK            13
10 Bedroom      12
9 BHK            8
8 BHK            5
```

```
11 BHK           2
11 Bedroom       2
10 BHK           2
14 BHK           1
13 BHK           1
12 Bedroom       1
27 BHK           1
43 Bedroom       1
16 BHK           1
19 BHK           1
18 Bedroom       1
Name: count, dtype: int64
--------------------------------
society
GrrvaGr    80
PrarePa    76
Sryalan    59
Prtates    59
GMown E    56
           ..
Amionce     1
JaghtDe     1
Jauraht     1
Brity U     1
RSntsAp     1
Name: count, Length: 2688, dtype: int64
--------------------------------
total_sqft
1200    843
1100    221
1500    205
2400    196
600     180
       ...
3580      1
2461      1
1437      1
2155      1
4689      1
Name: count, Length: 2117, dtype: int64
--------------------------------
bath
2.0     6908
3.0     3286
4.0     1226
1.0      788
5.0      524
6.0      273
7.0      102
8.0       64
9.0       43
10.0      13
12.0       7
13.0       3
11.0       3
16.0       2
```

```
27.0       1
40.0       1
15.0       1
14.0       1
18.0       1
Name: count, dtype: int64
--------------------------------
balcony
2.0    5113
1.0    4897
3.0    1672
0.0    1029
Name: count, dtype: int64
--------------------------------
price
75.00     310
65.00     302
55.00     275
60.00     270
45.00     240
         ...
351.00      1
54.10       1
80.64       1
32.73       1
488.00      1
Name: count, Length: 1994, dtype: int64
--------------------------------
```

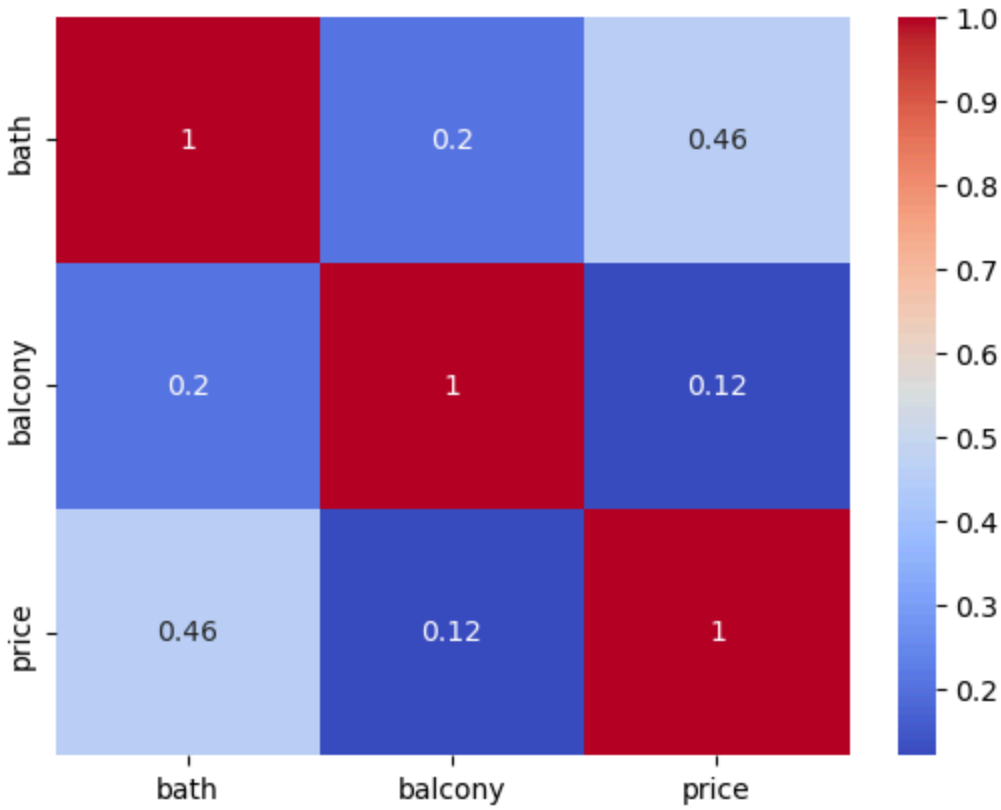
In [10]:
```
sns.pairplot(df)
```

Out[10]:
```
<seaborn.axisgrid.PairGrid at 0x1b637eba660>
```



In [11]:
```
num_vars = ["bath", "balcony", "price"]
sns.heatmap(df[num_vars].corr(),cmap="coolwarm", annot=True)
```

Out[11]: <Axes: >



Preare Data for Machine Learning Model

Data cleaning

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
area_type          0
availability       0
location           1
size              16
society         5502
total_sqft         0
bath              73
balcony          609
price              0
dtype: int64
```

In [13]:
```
df.isnull().mean()*100
```

Out[13]:
```
area_type        0.000000
availability     0.000000
location         0.007508
size             0.120120
society         41.306306
total_sqft       0.000000
bath             0.548048
balcony          4.572072
price            0.000000
dtype: float64
```

Remove 'society' due to high nullity rate (41%)

In [14]:
```
df2 = df.drop('society', axis='columns')
df2.shape
```

Out[14]:
```
(13320, 8)
```

Replace the null value in 'balcony' with the average of the non-null values.

In [15]:
```
df2['balcony'] = df2['balcony'].fillna(df2['balcony'].mean())
df2.isnull().sum()
```

Out[15]:
```
area_type        0
availability     0
location         1
size            16
total_sqft       0
bath            73
balcony          0
price            0
dtype: int64
```

Remove NaN

In [16]:
```
df3 = df2.dropna()
df3.shape
```

Out[16]:
```
(13246, 8)
```

In [17]:
```
df3.isnull().sum()
```

Out[17]:
```
area_type       0
availability    0
location        0
size            0
total_sqft      0
bath            0
balcony         0
price           0
dtype: int64
```

Feature Engineering

In [18]:
```
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", None)
```

1. Converting 'total_sqft' cat feature in numeric

In [19]:
```
df3['total_sqft'].value_counts()
```

Out[19]:

total_sqft	
1200	843
1100	221
1500	204
2400	195
600	180
1000	172
1350	132
1050	123
1300	117
1250	114
900	112
1400	108
1800	104
1150	101
1600	100
1140	91
2000	82
1450	70
1650	69
800	67
3000	66
1075	66
1020	63
2500	62
1125	60
1160	60
1550	60
950	59
1700	58
1180	58
1260	57
1255	56
1080	55
1220	55
1070	53
750	52
700	52
4000	48
1175	48
1225	48
1320	46
1240	46
2100	46
1230	45
1060	45
1210	44
850	43
1280	42
1270	41
1185	41
1190	40
1170	40
1410	40
1750	39
1025	38

1330	38
1310	37
1290	37
1850	37
1065	36
1194	36
1215	35
1090	35
500	34
1360	33
2700	33
1115	33
1900	32
3500	32
1120	32
1464	32
1530	31
2200	31
1205	31
1430	31
1340	31
1035	30
1165	30
1560	30
1130	29
1145	29
1128	29
3600	29
1355	28
1040	28
2800	28
1275	28
1105	27
1420	27
1155	27
1680	27
1760	25
1245	25
1216	25
1460	25
1590	25
650	25
1010	24
1475	24
1305	24
2600	24
1030	23
1110	23
1440	23
883	23
1575	22
1495	22
985	22
1246	22
1315	21
1610	21

1385	21
1027	21
1015	21
1470	21
1325	21
3200	21
1370	21
1243	21
660	20
1480	20
550	20
5000	20
925	20
1540	20
1390	20
1645	19
920	19
1365	19
1520	19
1265	19
1640	19
1570	19
1485	18
1095	18
4800	18
980	18
1525	18
1196	18
1345	18
1665	18
975	18
1195	18
1012	18
1820	18
2072	18
940	18
1375	17
960	17
400	17
1157	17
1295	17
1232	17
525	17
1045	16
2300	16
1830	16
1490	16
1418	16
1655	16
935	16
1135	16
3300	16
645	16
1197	16
1720	16
1235	16

1425	16
1153	15
1950	15
1660	15
1380	15
1116	15
4500	15
1740	15
1152	15
1085	15
1285	15
1920	15
1580	15
630	15
905	14
450	14
1630	14
1141	14
1445	14
720	14
675	14
1595	14
3900	13
3800	13
1875	13
1615	13
1082	13
1730	13
1535	13
840	13
1710	13
1690	13
1282	13
1510	13
1005	13
1314	13
1625	13
2215	13
970	13
2250	13
1033	12
1565	12
1404	12
1007	12
1296	12
2900	12
1151	12
1339	12
1691	12
1435	12
1890	12
1455	12
1056	12
1252	12
3100	12
1843	12

1162	11
1405	11
1174	11
1452	11
1515	11
845	11
1206	11
984	11
1804	11
1639	11
1346	11
540	11
1685	11
2150	11
965	11
1198	10
1256	10
3400	10
918	10
2350	10
1620	10
880	10
1415	10
995	10
1465	10
4200	10
1555	10
620	10
930	10
770	10
760	10
3596	10
1161	10
1063	10
957	10
1308	10
1810	10
1092	9
1605	9
1419	9
2180	9
780	9
1693	9
1224	9
2750	9
1715	9
910	9
1223	9
1724	9
6000	9
945	9
1186	9
1482	9
1725	9
3150	9
1113	9

1156	9
705	9
1047	9
3750	9
1756	9
890	9
2790	9
1790	9
1745	9
1870	9
2475	9
674	9
3250	9
825	9
1108	9
1222	9
5400	8
1133	8
875	8
1636	8
2990	8
1088	8
812	8
1335	8
1221	8
1178	8
1299	8
1705	8
1835	8
1146	8
656	8
1532	8
680	8
1395	8
1447	8
1352	8
1187	8
1163	8
1476	8
1026	8
1703	8
870	8
2145	8
1089	8
1322	8
990	8
1192	8
1670	8
2650	8
4400	8
3122	8
1276	7
1101	7
1139	7
1044	7
1166	7

1718	7
1183	7
1397	7
820	7
1093	7
1357	7
1112	7
1041	7
1096	7
1374	7
1545	7
1427	7
1306	7
1132	7
1262	7
1564	7
1840	7
1277	7
982	7
1272	7
1251	7
2050	7
1762	7
1254	7
1508	7
1199	7
708	7
1917	7
1910	7
1179	7
615	7
1880	7
1392	7
7500	7
1143	7
1847	7
1349	7
1069	7
1342	7
1036	7
1213	6
440	6
530	6
1278	6
1614	6
1846	6
2480	6
560	6
1787	6
1635	6
1173	6
625	6
924	6
1303	6
1362	6
1484	6

1697	6
1453	6
1084	6
1242	6
1053	6
1780	6
1735	6
710	6
7000	6
2040	6
1016	6
1583	6
605	6
2425	6
1411	6
1755	6
1304	6
1107	6
2010	6
1274	6
1675	6
933	6
1719	6
3895	6
1247	6
860	6
830	6
1541	6
1893	6
1785	6
1512	6
1167	6
520	6
654	6
1181	6
1451	6
2690	6
1257	6
610	6
1936	6
2275	6
1571	6
1856	6
1268	6
1313	6
936	6
2774	6
1127	6
1333	6
1372	6
1656	6
1176	6
1602	6
1770	6
1991	6
1286	6

1028	6
1717	6
1855	6
810	6
929	6
1083	6
1019	6
1062	6
1102	6
2280	6
2225	6
1126	6
1329	5
1865	5
1052	5
1328	5
919	5
1123	5
1021	5
2710	5
1263	5
1884	5
1344	5
4395	5
4050	5
1076	5
1754	5
1801	5
1505	5
545	5
1226	5
1293	5
1868	5
1664	5
1253	5
1168	5
1244	5
915	5
1403	5
1662	5
907	5
1297	5
2254	5
410	5
1826	5
1309	5
1559	5
2850	5
1267	5
1326	5
1798	5
1738	5
1057	5
1930	5
1654	5
1077	5

1009	5
1533	5
1862	5
1148	5
1573	5
420	5
2360	5
635	5
1586	5
2830 - 2882	5
1055	5
1022	5
865	5
1358	5
1408	5
1765	5
1935	5
510	5
1683	5
1852	5
1098	5
715	5
1825	5
1458	5
1184	5
1188	5
1208	5
1749	5
3450	5
1231	5
2119	5
740	5
1436	5
1204	5
1351	5
640	5
1142	5
1354	5
1925	5
1567	5
1364	5
1457	5
1444	5
1031	5
1585	5
1109	5
1269	5
2340	5
1424	5
1241	5
1980	4
1702	4
1521	4
1904	4
993	4
2560	4

1072	4
662	4
1046	4
1711	4
946	4
1842	4
1837	4
1449	4
1466	4
1067	4
1037	4
2160	4
460	4
1259	4
418	4
2070	4
1864	4
1238	4
1111	4
1965	4
921	4
1091	4
1394	4
955	4
755	4
972	4
1367	4
3520	4
1494	4
1202	4
1929	4
3850	4
1537	4
1708	4
1752.12	4
2830	4
8000	4
1169	4
2640	4
966	4
1695	4
2805	4
1795	4
1343	4
1219	4
992	4
3700	4
1353	4
4750	4
1938	4
1081	4
1034	4
1171	4
814	4
1201	4
1258	4

1158	4
1291	4
877	4
1059	4
2450	4
360	4
927	4
1698	4
914	4
1616	4
1976	4
1386	4
1608	4
2439	4
1891	4
996	4
1061	4
1885	4
3436	4
1732	4
1611	4
1236	4
2240	4
1332	4
1945	4
1008	4
12000	4
2135	4
967	4
1933	4
1073	4
1767	4
497	4
595	4
1118	4
1318	4
902	4
1975	4
602	4
795	4
861	4
1563	4
10000	4
1234	4
1776	4
1203	4
1134	4
1074	4
1407	4
1432	4
665	4
1051	4
416	4
4100	4
1504	4
1517	4

1592	4
1138	4
1634	4
2273	4
1058	4
1596	4
1279	4
1489	4
2760	4
1806	4
1334	4
775	4
580	4
884	4
1104	4
1207	4
1881	4
1079	4
1459	4
1149	4
1569	4
1576	3
3730	3
648	3
375	3
1398	3
1817	3
1382	3
1086	3
7200	3
1588	3
2559	3
1896	3
2106	3
829	3
2230	3
664	3
1594	3
994	3
2017	3
3385	3
1301	3
3155	3
1011	3
745	3
1307	3
702	3
1527	3
1147	3
896	3
1858	3
1172	3
1336	3
923	3
1553	3
891	3

1478	3
782	3
9600	3
1839	3
1845	3
2062	3
1513	3
1182	3
1154	3
1903	3
1788	3
1144	3
1603	3
1692	3
1066	3
2503	3
1311	3
1768	3
1942	3
2060	3
1448	3
2357	3
2440	3
1097	3
1018	3
2610	3
1212	3
1653	3
1531	3
1536	3
1819	3
1584	3
2065	3
2940	3
1682	3
1426	3
6200	3
1428	3
1591	3
1124	3
2502	3
1672	3
1043	3
703	3
1164	3
3067	3
2257	3
2289	3
2770	3
3526	3
1302	3
1071	3
2095	3
1599	3
2020	3
2080	3

937	3
1566	3
1577	3
1960	3
5100	3
2880	3
2732	3
1094	3
670	3
1191	3
672	3
2087.01	3
1006	3
2325	3
658	3
1984	3
1860	3
2390	3
3040	3
1294	3
690	3
2292	3
4600	3
2470	3
1347	3
2550	3
1491	3
1689	3
1832	3
1017	3
1726	3
1737	3
1312	3
912	3
1783	3
999	3
6500	3
1477	3
1519	3
663	3
1777.26	3
425	3
1704	3
1694	3
1209	3
2330	3
2144	3
4111	3
3205	3
1121	3
480	3
1518	3
1416	3
1341	3
1706	3
1713	3

1384	3
1496	3
1853	3
1356	3
2882	3
1233	3
1779	3
3252	3
1652	3
1024	3
3630	3
1758	3
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1897	3
958	3
1644	3
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1371	2
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952	2
3930	2

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1613	2
1284	2
1677	2
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2170	2
1137	2
1676	2
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2041	2
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3170	2
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876	2
1107.83	2
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827	2
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596	2

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938	2
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1914	2

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1396	2
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1323	2
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2045	2
585	2
765	2

2025	2
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1229	2
2089	2
3761	2
2524	2
986	2
1115 - 1130	2
5500	2
1499	2
1388	2
645 - 936	2
686	2
2238	2
1995	2
509	2
2556	2
793	2
1001	2
1554	2
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913	2
726	2
1556	2
1808	2
3024	2
1804 - 2273	2
1552	2
551	2
1542	2
805	2
2409	2
1239	2
3198	2
3356	2
589	2
6652	2
1612	2
1684	2
1359	2
1739	2
1381	2
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2197	2
4346	2
1363	2
1524	2

2006	2
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682	2
5800	2
1950.2	2
1666	2
2420	2
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1937	2
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2689	2
1651	2
142.61Sq. Meter	2
1078	2
1757	2
882	2
10961	2
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834	2
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2321	2
527	2
606	2
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2130	1
2955	1
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300Sq. Yards	1
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2646	1
1686	1
1989	1
1100Sq. Meter	1
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666	1
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754	1
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2736	1
616	1
4000 - 4450	1
3056	1
1998	1
142.84Sq. Meter	1
2999.97	1
3293	1
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4110	1

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1506	1
772	1
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2295	1
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3230	1
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3290	1
596 - 861	1
5656	1
1962	1
3746	1
1542.14	1
567	1
1400 - 1421	1
1568	1
4900 - 4940	1
850 - 1060	1
5422	1
3060	1
873	1
1.26Acres	1
4350	1
1443	1
650 - 760	1
1623.29	1
755 - 770	1
475	1
3307 - 3464	1

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4723	1
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540 - 740	1
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2485	1
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5200	1
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20000	1
3508 - 4201	1
1437 - 1629	1
6830	1
3692	1
2505	1
866	1
1503	1
1273	1
3978	1
763 - 805	1
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120Sq. Yards	1
30400	1
3870	1
1783 - 1878	1
4097	1
664 - 722	1
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2380	1
712 - 938	1
2206	1
3569	1
151.11Sq. Yards	1
3435	1
3301.8	1
2444	1
3144	1
3602	1
1857	1
953	1
1502	1
5515	1
4166	1
2008	1
886	1
694	1
2031	1
3033	1
2086	1

2528 - 3188	1
1181.7	1
16335	1
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3489	1
1932.47	1
1338	1
2625	1
1317	1
1321	1
747	1
24Sq. Meter	1
2153	1
1589	1
7150	1
4041	1
2470 - 2790	1
4482	1
783 - 943	1
1189	1
1905	1
1940	1
5384	1
1723	1
5150	1
1990	1
2956	1
250	1
1522	1
4303	1
3210	1
1626.6	1
2316	1
1383	1
516	1
581	1
5425	1
2088	1
2916	1
978	1
488	1
4446	1
1150 - 1194	1
613	1
2223	1
9200	1
1248.52	1
1604	1
717	1
1748	1
1510 - 1670	1
3019	1
3535	1
3117	1
2648	1
1761	1

2098	1
1907	1
45.06Sq. Meter	1
3410	1
799 - 803	1
2185	1
998	1
866.28	1
2733	1
1389	1
2970	1
684 - 810	1
2507	1
2064	1
3628	1
1547	1
896.9	1
1287	1
1361	1
2735	1
86.72Sq. Meter	1
2801.25	1
1.25Acres	1
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1688.12	1
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2650 - 2990	1
3350	1
2758	1
1255 - 1375	1
1981	1
1733.5	1
10624	1
3418	1
691	1
3092	1
2597	1
854 - 960	1
610 - 615	1
3913	1
1049	1
2404	1
1642	1
2285	1
284	1
4550	1
2082	1
2395	1
3075	1
2437	1
3270	1
2495	1
857	1
1230 - 1490	1
524	1
2247	1

2980	1
2826	1
1133 - 1384	1
2087	1
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499	1
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1731	1
2415	1
1609	1
1861	1
4209	1
3362	1
4304	1
1331.95	1
1140 - 1250	1
6Acres	1
842	1
1079 - 1183	1
2800 - 2870	1
2015	1
1230 - 1290	1
2048	1
1215 - 1495	1
1119	1
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2429	1
2406	1
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1747	1
2019	1
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2162.03	1
614	1
5666 - 5669	1

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2504	1
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2400 - 2600	1
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750 - 800	1
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2704	1
3760	1
2035	1
1052 - 1322	1
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2787	1
2137	1
2872	1
2695	1
943 - 1220	1
3496	1
3381	1
706 - 716	1
3560	1
1450 - 1950	1
1379	1
2150 - 2225	1
2940Sq. Yards	1
929 - 1078	1
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774	1
612	1
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623	1
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2312	1
1797	1
976	1
1816	1
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629 - 1026	1
1648	1
302	1
2041 - 2090	1
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1234.6	1
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527 - 639	1
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534	1
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1520 - 1759	1
2110	1
4560	1
660 - 780	1
1070 - 1315	1
2Acres	1
2611	1
1005.03 - 1252.49	1
1901	1
3245	1
515	1
3025	1
1004 - 1204	1
361.33Sq. Yards	1
987	1
2121	1
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4850	1
668	1
2710 - 3360	1
395	1
1452.55	1
296	1
714	1
351	1
2526	1
596 - 804	1
1776.42	1
2319	1
11338	1
30000	1
3190	1
4460	1
1316	1
2039	1
983	1
2172.65	1
1722	1
1429	1
1574Sq. Yards	1
3606	1
2337	1
5080	1

52272	1
2479.13	1
3042	1
2204	1
916	1
688	1
3555	1
1687	1
6150	1
671	1
581.91	1
1250 - 1305	1
824	1
2519	1
1673	1
670 - 980	1
3584	1
2168	1
3125	1
1775	1
3297	1
1255 - 1863	1
1300 - 1405	1
1451.5	1
3640	1
500Sq. Yards	1
2043	1
2779	1
1020.07	1
5965	1
2806 - 3019	1
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956	1
1218	1
2363	1
885	1
3016	1
1888	1
673	1
704 - 730	1
2408	1
685	1
3530	1
3950	1
1205.47	1
3595	1
944	1
832	1
3259	1
858	1
3040Sq. Meter	1
2220	1
590	1
980 - 1030	1
3161	1
871	1

3515	1
3366	1
1562	1
2105	1
2167	1
117Sq. Yards	1
904	1
977	1
2401	1
934 - 1437	1
1452.19	1
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3010	1
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792	1
461.82	1
989	1
1818	1
2820	1
4772	1
5600	1
4051	1
2249.81 - 4112.19	1
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727	1
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1500Sq. Meter	1
1753	1
1511	1
8500	1
5230	1
1867	1
2845	1
1195 - 1440	1
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961	1
2791	1
469	1
4125Perch	1
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1120 - 1145	1
3090 - 5002	1
35000	1
2118	1
1337	1
5270	1
667	1
4360	1
1160 - 1195	1
1000Sq. Meter	1
445	1
2181	1
3670	1
1114	1
3309	1

11890	1
34.46Sq. Meter	1
1563.05	1
2785	1
2100 - 2850	1
1330.74	1
869	1
5700	1
2511	1
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1413	1
2497	1
276	1
3067 - 8156	1
1042 - 1105	1
1828	1
1113.27	1
2795	1
1145 - 1340	1
1015 - 1540	1
1909	1
1736	1
2405	1
3045	1
2026	1
11	1
2023.71	1
451	1
1629	1
2489	1
1100Sq. Yards	1
520 - 645	1
3680	1
3329	1
785	1
716Sq. Meter	1
3073	1
764	1
340	1
2465	1
547.34 - 827.31	1
628	1
804.1	1
580 - 650	1
1373	1
2863	1
2195	1
2424	1
3516	1
3770	1
1378	1
2572	1
3425 - 3435	1
1269.72	1
1266	1
817	1

1815	1
4000 - 5249	1
1193	1
2090	1
3467.86	1
1000 - 1285	1
2557	1
4239	1
2825	1
650 - 665	1
633 - 666	1
315	1
2112.95	1
5.31Acres	1
15	1
1467	1
3968	1
2169	1
3235	1
30Acres	1
4634	1
1824	1
1561	1
1637	1
24	1
1445 - 1455	1
884 - 1116	1
850 - 1093	1
14000	1
1617	1
2842	1
1558.67	1
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1689.28	1
2912	1
2122	1
1525.84	1
2563 - 2733	1
2005	1
605 - 624	1
2582	1
2663	1
2127	1
1349 - 3324	1
45	1
78.03Sq. Meter	1
1208.51	1
3300 - 3335	1
2435	1
2328	1
1180 - 1630	1
806	1
1628	1
1660.4	1
4075	1
1987	1

2245	1
1597	1
2423	1
3131	1
396	1
4273	1
122Sq. Yards	1
934	1
941	1
2384	1
906	1
1014	1
797	1
2396	1
1939	1
3401	1
1469 - 1766	1
2246	1
4510	1
204Sq. Meter	1
1766	1
870 - 1080	1
45Sq. Yards	1
948	1
695	1
2601	1
3160	1
1769	1
133.3Sq. Yards	1
1554.3	1
4920	1
627	1
3405.1	1
2570	1
881	1
951	1
1652.5	1
3532	1
84.53Sq. Meter	1
651	1
2.09Acres	1
1412	1
790	1
2476	1
1877	1
1649	1
766	1
1593	1
598 - 958	1
1631	1
1500Cents	1
1829	1
5720	1
2132	1
1439	1
2679	1

5924	1
552	1
2365	1
864	1
2293	1
1567.2	1
26136	1
959	1
132Sq. Yards	1
855	1
1691.2	1
1010 - 1300	1
3044	1
942 - 1117	1
583	1
697Sq. Meter	1
981 - 1249	1
2108	1
2968	1
1565 - 1595	1
2232	1
2932	1
3820	1
24Guntha	1
1831	1
1270 - 1275	1
840 - 1010	1
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1266.67	1
638	1
4856	1
3664	1
539	1
2173	1
655 - 742	1
1741	1
3463	1
2120	1
505	1
1408 - 1455	1
2901	1
188.89Sq. Yards	1
2302	1
2059	1
1548.3	1
2342	1
6600	1
1627.86	1
1874	1
2885	1
1462	1
607	1
1053.4	1
540 - 670	1
315Sq. Yards	1

3445	1
2283	1
3815	1
4827	1
833	1
669	1
3811	1
3860	1
3027	1
1734	1
704	1
2725 - 3250	1
1732.46	1
2592	1
4470	1
624	1
722	1
2270	1
2651	1
1076 - 1199	1
4818	1
888	1
532	1
3227	1
1915	1
963	1
769	1
1712	1
1918	1
2113	1
2386	1
1210 - 1477	1
892	1
4355	1
167Sq. Meter	1
908	1
3369 - 3464	1
1574	1
1125 - 1500	1
813	1
3884	1
3876	1
897	1
2776	1
3480	1
1	1
2372	1
2775	1
706	1
4356	1
2792	1
3080	1
787	1
660 - 700	1
6613	1
502	1

```
3339                    1
1483                    1
2162                    1
1733                    1
1974                    1
385 - 440               1
2631                    1
2448                    1
4830                    1
770 - 841               1
2171                    1
42000                   1
1793                    1
1551                    1
3563                    1
462                     1
3Cents                  1
36000                   1
1113.12                 1
621                     1
981                     1
2383                    1
555                     1
2138                    1
9000                    1
910.2                   1
4960                    1
3179                    1
2928                    1
1932                    1
2182                    1
2416                    1
3905                    1
2107                    1
2431                    1
4640                    1
818                     1
1288                    1
2155                    1
300                     1
1778                    1
2006.8                  1
1922                    1
888 - 1290              1
2370                    1
5                       1
1004                    1
5985                    1
3580                    1
2461                    1
1437                    1
4689                    1
Name: count, dtype: int64
```

2. Converting total_sqft to float

In [20]:
```
total_sqft_float = []
for str_val in df3['total_sqft']:
  try:
    total_sqft_float.append(float(str_val))
  except:
    try:
      temp = []
      temp = str_val.split('-')
      total_sqft_float.append((float(temp[0])+float(temp[-1]))/2)
    except:
      total_sqft_float.append(np.nan)
```

In [21]:
```
df4 = df3.reset_index(drop=True)
```

In [22]:
```
df5 = df4.join(pd.DataFrame({'total_sqft_float':total_sqft_float}))
df5.head()
```

Out[22]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	tc
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	1056	2.0	1.0	39.07	
1	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	2600	5.0	3.0	120.00	
2	Built-up Area	Ready To Move	Uttarahalli	3 BHK	1440	2.0	3.0	62.00	
3	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	1521	3.0	1.0	95.00	
4	Super built-up Area	Ready To Move	Kothanur	2 BHK	1200	2.0	1.0	51.00	

In [23]:
```
df5.isnull().sum()
```

Out[23]:
```
area_type           0
availability        0
location            0
size                0
total_sqft          0
bath                0
balcony             0
price               0
total_sqft_float   46
dtype: int64
```

In [24]:
```
df6 = df5.dropna()
df6.shape
```

Out[24]: (13200, 9)

In [25]:
```
df6.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 13200 entries, 0 to 13245
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   area_type         13200 non-null  object 
 1   availability      13200 non-null  object 
 2   location          13200 non-null  object 
 3   size              13200 non-null  object 
 4   total_sqft        13200 non-null  object 
 5   bath              13200 non-null  float64
 6   balcony           13200 non-null  float64
 7   price             13200 non-null  float64
 8   total_sqft_float  13200 non-null  float64
dtypes: float64(4), object(5)
memory usage: 1.0+ MB
```

3. Working with feature: *size*

In [26]:
```
df6['size'].value_counts()
```

Out[26]:
```
size
2 BHK         5192
3 BHK         4277
4 Bedroom      816
4 BHK          574
3 Bedroom      541
1 BHK          527
2 Bedroom      325
5 Bedroom      293
6 Bedroom      190
1 Bedroom      100
7 Bedroom       83
8 Bedroom       83
5 BHK           56
9 Bedroom       45
6 BHK           30
7 BHK           17
1 RK            13
10 Bedroom      12
9 BHK            7
8 BHK            5
11 BHK           2
11 Bedroom       2
10 BHK           2
14 BHK           1
13 BHK           1
12 Bedroom       1
27 BHK           1
43 Bedroom       1
16 BHK           1
19 BHK           1
18 Bedroom       1
Name: count, dtype: int64
```

In [27]:
```
size_int = []
for str_val in df6['size']:
  temp=[]
  temp = str_val.split(" ")
  try:
    size_int.append(int(temp[0]))
  except:
    size_int.append(np.nan)
    print("Noice = ",str_val)
```

In [28]:
```
df6 = df6.reset_index(drop=True)
```

In [29]:
```
df7 = df6.join(pd.DataFrame({'bhk':size_int}))
df7.shape
```

Out[29]:
```
(13200, 10)
```

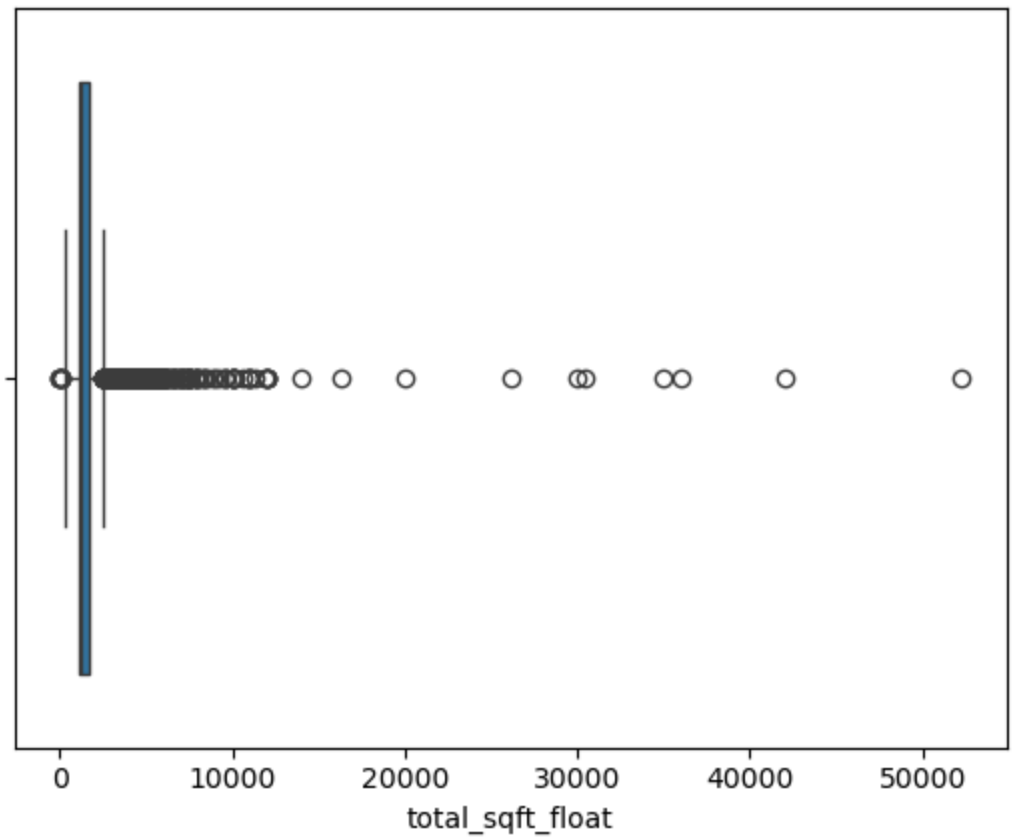
In [30]:
```
df7.tail()
```

Out[30]:

	area_type	availability	location	size	total_sqft	bath	balcony	price
13195	Built-up Area	Ready To Move	Whitefield	5 Bedroom	3453	4.0	0.000000	231.0
13196	Super built-up Area	Ready To Move	Richards Town	4 BHK	3600	5.0	1.584376	400.0
13197	Built-up Area	Ready To Move	Raja Rajeshwari Nagar	2 BHK	1141	2.0	1.000000	60.0
13198	Super built-up Area	18-Jun	Padmanabhanagar	4 BHK	4689	4.0	1.000000	488.0
13199	Super built-up Area	Ready To Move	Doddathoguru	1 BHK	550	1.0	1.000000	17.0

In [31]:
```
sns.boxplot(x = df7['total_sqft_float'])
```

Out[31]:
```
<Axes: xlabel='total_sqft_float'>
```



In [32]:
```
df7[df7['total_sqft_float']/df7['bhk'] < 350].head()
```

Out[32]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	tota
9	Plot Area	Ready To Move	Gandhi Bazar	6 Bedroom	1020	6.0	1.584376	370.0	
26	Super built-up Area	Ready To Move	Electronic City	2 BHK	660	1.0	1.000000	23.1	
29	Super built-up Area	Ready To Move	Electronic City	3 BHK	1025	2.0	1.000000	47.0	
45	Plot Area	Ready To Move	HSR Layout	8 Bedroom	600	9.0	1.584376	200.0	
57	Plot Area	Ready To Move	Murugeshpalya	6 Bedroom	1407	4.0	1.000000	150.0	

In [33]:
```
df8 = df7[~(df7['total_sqft_float']/df7['bhk'] < 350)]
df8.shape
```

Out[33]:
```
(12106, 10)
```

In [34]:
```
df8['price_per_sqft'] = df8['price']*100000 / df8['total_sqft_float']
df8.head()
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_33656\1116913982.py:1: SettingWithCopyWa
rning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  df8['price_per_sqft'] = df8['price']*100000 / df8['total_sqft_float']
```

Out[34]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	tc
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	1056	2.0	1.0	39.07	
1	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	2600	5.0	3.0	120.00	
2	Built-up Area	Ready To Move	Uttarahalli	3 BHK	1440	2.0	3.0	62.00	
3	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	1521	3.0	1.0	95.00	
4	Super built-up Area	Ready To Move	Kothanur	2 BHK	1200	2.0	1.0	51.00	

In [35]:
```
df8.price_per_sqft.describe()
```

Out[35]:
```
count     12106.000000
mean       6184.466889
std        4019.983503
min         267.829813
25%        4200.030048
50%        5261.108523
75%        6800.000000
max      176470.588235
Name: price_per_sqft, dtype: float64
```

Homework

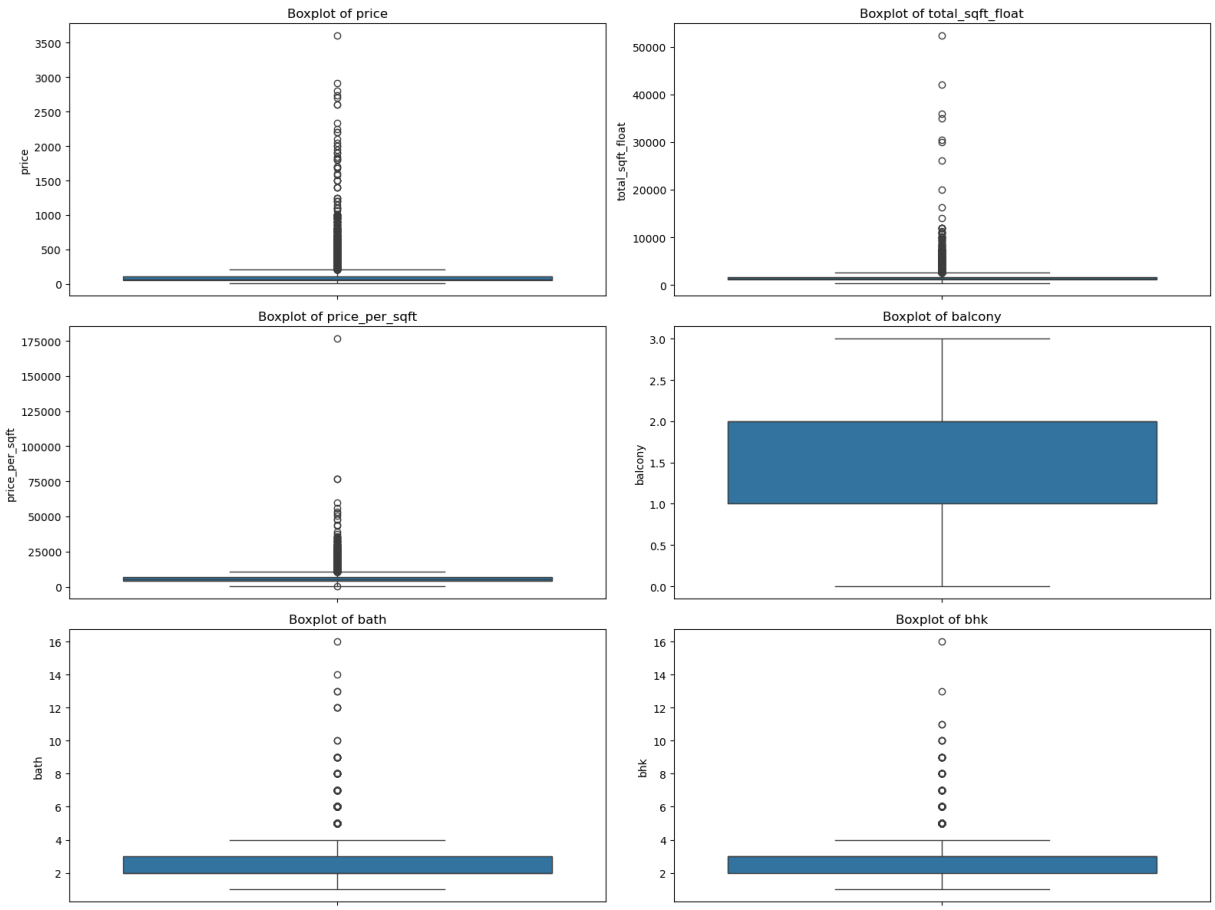
Bài tập 0: Sử dụng sns.boxplot() để quan sát đặc điểm phân bố dữ liệu của các trường số, mỗi trường này có outlier ko?

In [36]:
```
# Sử dụng boxplot để quan sát phân bố của dữ liệu và phát hiện ngoại lai của từng t
# Gợi ý: sns.boxplot(data_field)

vars = ['price', 'total_sqft_float', 'price_per_sqft', 'balcony', 'bath', 'bhk']
plt.figure(figsize=(16,12))

for i, var in enumerate(vars):
    plt.subplot(3, 2, i+1)
    sns.boxplot(data=df8, y=var)
    plt.title(f'Boxplot of {var}')

plt.tight_layout()
plt.show()
```



Bài tập 1: Viết hàm bỏ đi các điểm dữ liệu có price per sqft dựa trên mean, std của các ngôi nhà dựa trên từng vị trí

> Gợi ý: Xét trên từng vị trí (location), ngôi nhà thỏa mãn phải có $price_per_sqft \in [mean - std, mean + std]$

In [38]:
```
def remove_pps_outliers(df):
    df_out = pd.DataFrame()
    for key, subdf in df.groupby('location'):
        m = np.mean(subdf.price_per_sqft)
        st = np.std(subdf.price_per_sqft)
        reduced_df = subdf[(subdf.price_per_sqft>(m-st)) & (subdf.price_per_sqft<=(
        df_out = pd.concat([df_out,reduced_df],ignore_index=True)
    return df_out

df9 = remove_pps_outliers(df8)
df9.shape
```

Out[38]: (8888, 11)

Bài tập 2: Loại bỏ outlier xét theo trường bkh (số phòng)

Xét theo từng khu vực địa lí và theo từng loại nhà với số lượng phòng khác nhau, có một số ngôi nhà có giá không hợp lí (outliers), hãy tìm cách loại bỏ các outlier này. Cần ghi rõ quy tắc ghi nhận outlier

In [39]:
```
def remove_bhk_outliers(df):
    exclude_indices = np.array([])
    for location, location_df in df.groupby('location'):
        bhk_stats = {}
        for bhk, bhk_df in location_df.groupby('bhk'):
            bhk_stats[bhk] = {
                'mean': np.mean(bhk_df.price_per_sqft),
                'std': np.std(bhk_df.price_per_sqft),
                'count': bhk_df.shape[0]
            }
        for bhk, bhk_df in location_df.groupby('bhk'):
            stats = bhk_stats.get(bhk-1)
            if stats and stats['count']>5:
                exclude_indices = np.append(exclude_indices, bhk_df[bhk_df.price_pe
    return df.drop(exclude_indices,axis='index')

df10 = remove_bhk_outliers(df9)
df10.shape
```

Out[39]: (7194, 11)

Bài tập 3: Loại bỏ outlier khi xét trường 'bathroom'

In [40]:
```
df10.bath.unique() #Có thể quan sát thấy một số căn nhà có số phòng tắm quá lớn (VD
```

Out[40]: array([3., 2., 1., 4., 5., 8., 9., 6., 7., 12.])

In [41]:
```
df10[df10.bath > df10.bhk+2]
```

Out[41]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	t
1861	Built-up Area	Ready To Move	Chikkabanavar	4 Bedroom	2460	7.0	2.000000	80.0	
5836	Built-up Area	Ready To Move	Nagasandra	4 Bedroom	7000	8.0	1.584376	450.0	
7098	Super built-up Area	Ready To Move	Sathya Sai Layout	6 BHK	11338	9.0	1.000000	1000.0	
7569	Super built-up Area	Ready To Move	Thanisandra	3 BHK	1806	6.0	2.000000	116.0	

In [42]:
```
df11 = df10[df10.bath < df10.bhk+2]
df11.shape
```

Out[42]: (7120, 11)

In [43]:
```
df11.head()
```

Out[43]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	to
0	Super built-up Area	Ready To Move	Devarabeesana Halli	3 BHK	1672	3.0	2.0	150.0	
1	Built-up Area	Ready To Move	Devarabeesana Halli	3 BHK	1750	3.0	3.0	149.0	
2	Super built-up Area	Ready To Move	Devarabeesana Halli	3 BHK	1750	3.0	2.0	150.0	
4	Super built-up Area	Ready To Move	Devarachikkanahalli	2 BHK	1250	2.0	2.0	40.0	
5	Plot Area	Ready To Move	Devarachikkanahalli	2 Bedroom	1200	2.0	2.0	83.0	

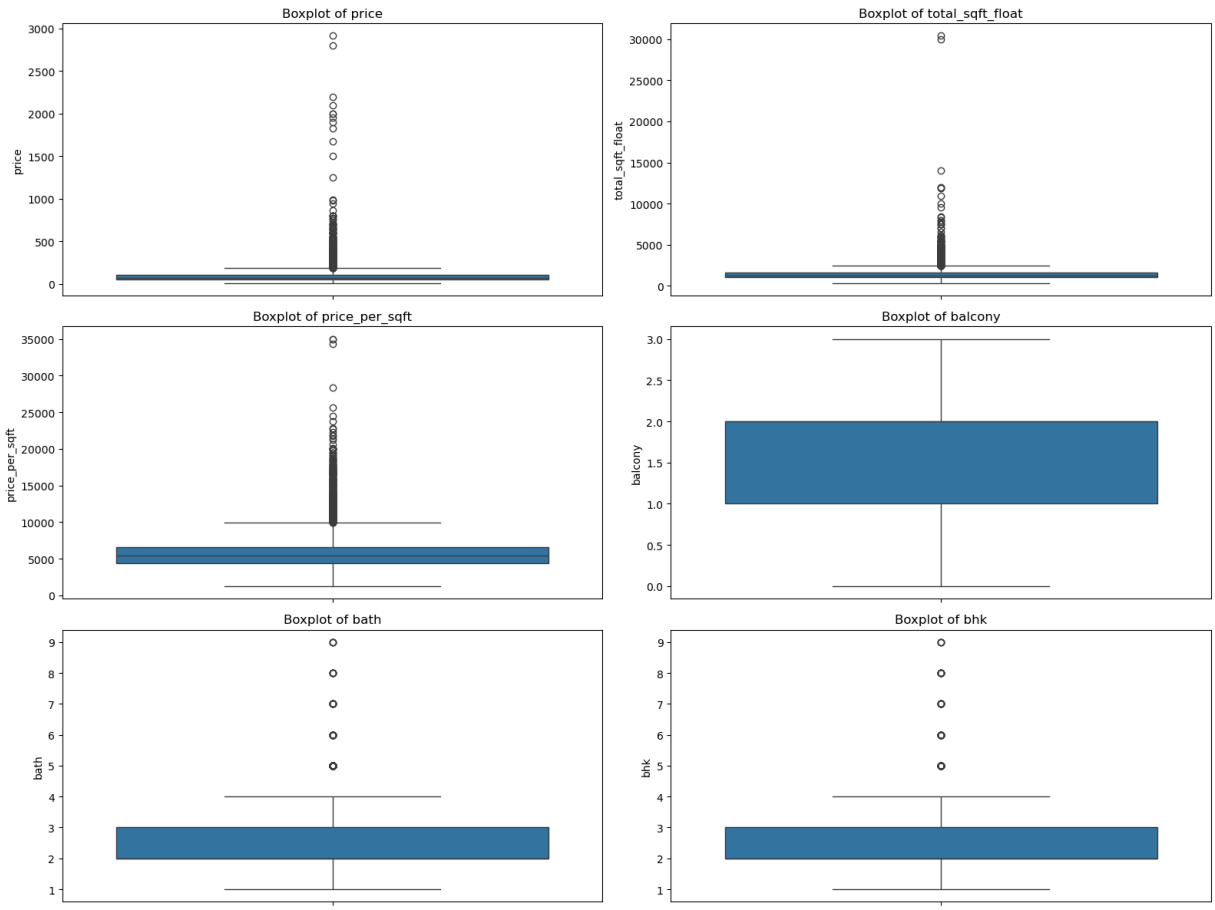
In [45]:

```
# Quan sát lại kết quả sau khi xử lí với boxplot

# (Dùng lại hàm đã code bên trên)
vars = ['price', 'total_sqft_float', 'price_per_sqft', 'balcony', 'bath', 'bhk']
plt.figure(figsize=(16,12))

for i, var in enumerate(vars):
    plt.subplot(3, 2, i+1)
    sns.boxplot(data=df11, y=var)
    plt.title(f'Boxplot of {var}')

plt.tight_layout()
plt.show()
```



Bài tập 4: Xem xét bỏ đi các trường không cần thiết

Gợi ý: bỏ đi ['area_type', 'availability',"location","size","total_sqft"]

In [46]:
```
df12 = df11.drop(['area_type', 'availability',"location","size","total_sqft"], axis
df12.head()
```

Out[46]:

	bath	balcony	price	total_sqft_float	bhk	price_per_sqft
0	3.0	2.0	150.0	1672.0	3	8971.291866
1	3.0	3.0	149.0	1750.0	3	8514.285714
2	3.0	2.0	150.0	1750.0	3	8571.428571
4	2.0	2.0	40.0	1250.0	2	3200.000000
5	2.0	2.0	83.0	1200.0	2	6916.666667

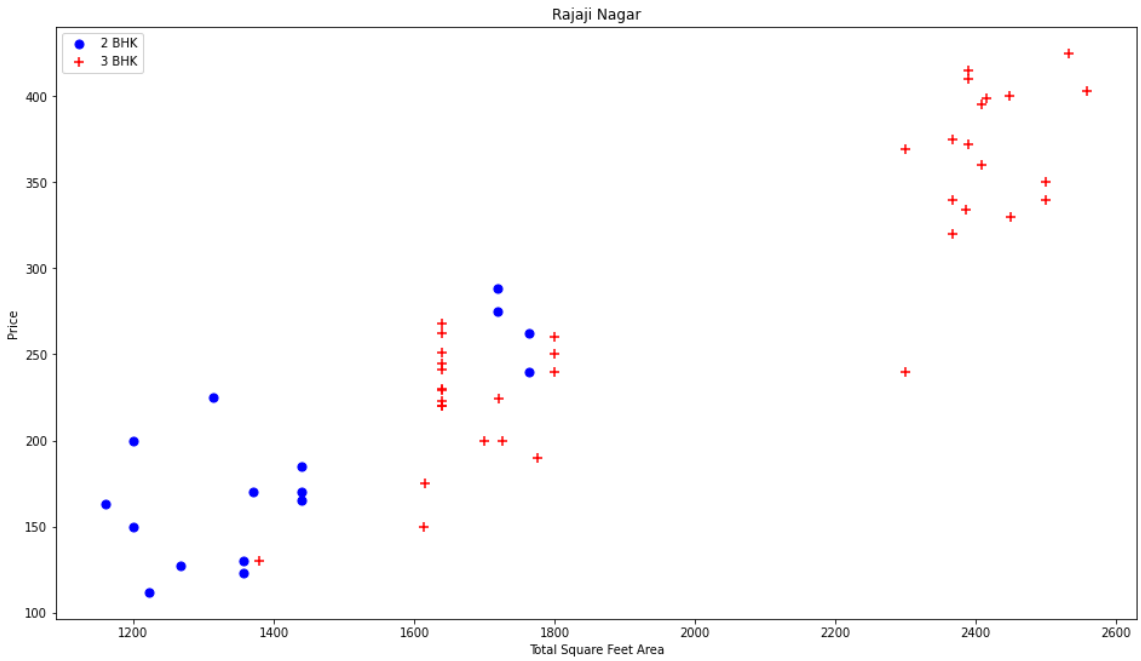
In [48]:
```
#Lưu kết quả xử lí cuối cùng:

df12.to_csv("clean_data2.csv", index=False)
```

Bài tập 5*: Viết hàm trực quan hóa thể hiện mối tương quan giữa tổng diện tích (total_sqft) và giá nhà (price) theo từng vị trí địa lí (location) (tùy chọn minh họa theo 2 vị trí nào đó),

của những căn nhà có 2 hoặc 3 phòng. Và cần phân biệt rõ điểm dữ liệu nào tương ứng với nhà có 2 phòng, điểm nào tương ứng với nhà có 3 phòng?

Gợi ý: Kết quả tương tự như hình dưới/ hoặc biểu đồ khác có ý nghĩa tương đương

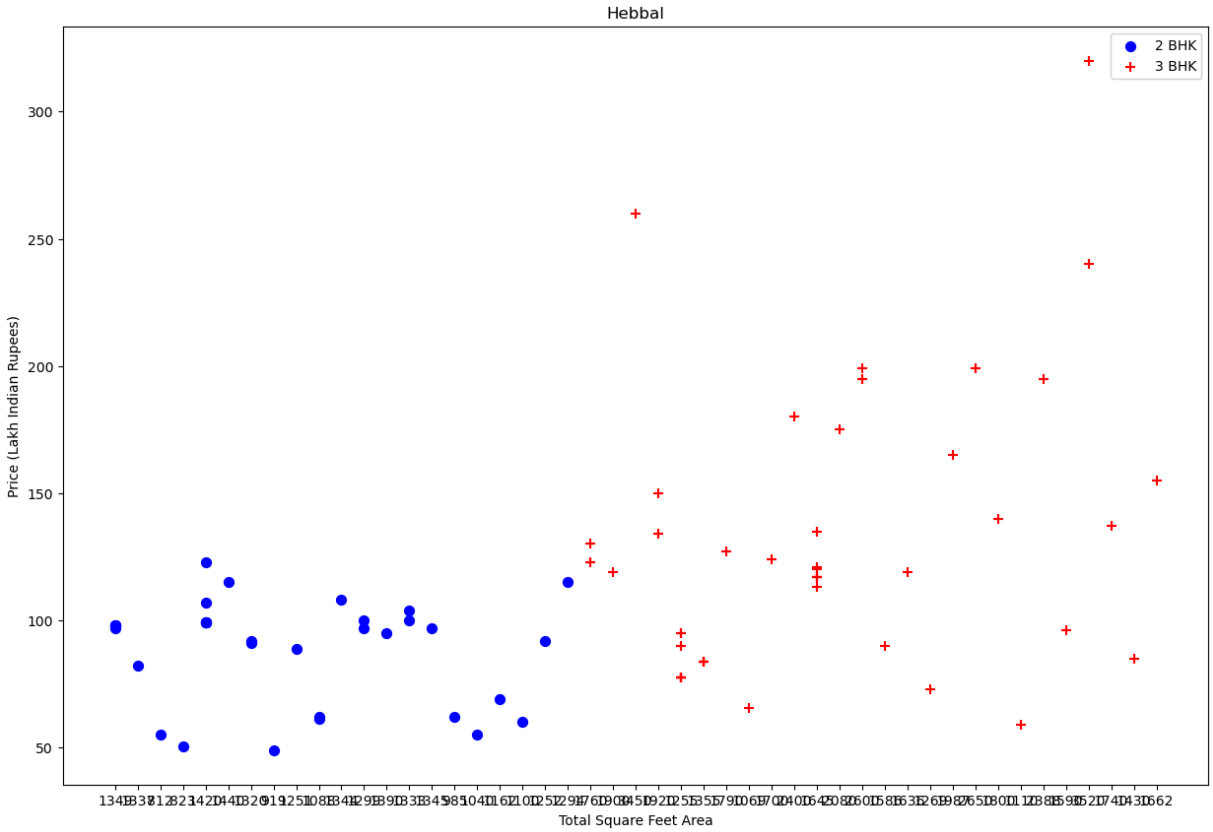


In [55]:
```
#Gợi ý: Sử dụng plt.scatter() .... hoặc câu lệnh khác tương đương. Làm với df9

import matplotlib


def plot_scatter_chart(df,location):
    bhk2 = df[(df.location==location) & (df.bhk==2)]
    bhk3 = df[(df.location==location) & (df.bhk==3)]
    matplotlib.rcParams['figure.figsize'] = (15,10)
    plt.scatter(bhk2.total_sqft,bhk2.price,color='blue',label='2 BHK', s=50)
    plt.scatter(bhk3.total_sqft,bhk3.price,color='red',label='3 BHK',s=50, marker='
    plt.xlabel("Total Square Feet Area")
    plt.ylabel("Price (Lakh Indian Rupees)")
    plt.title(location)
    plt.legend()
```

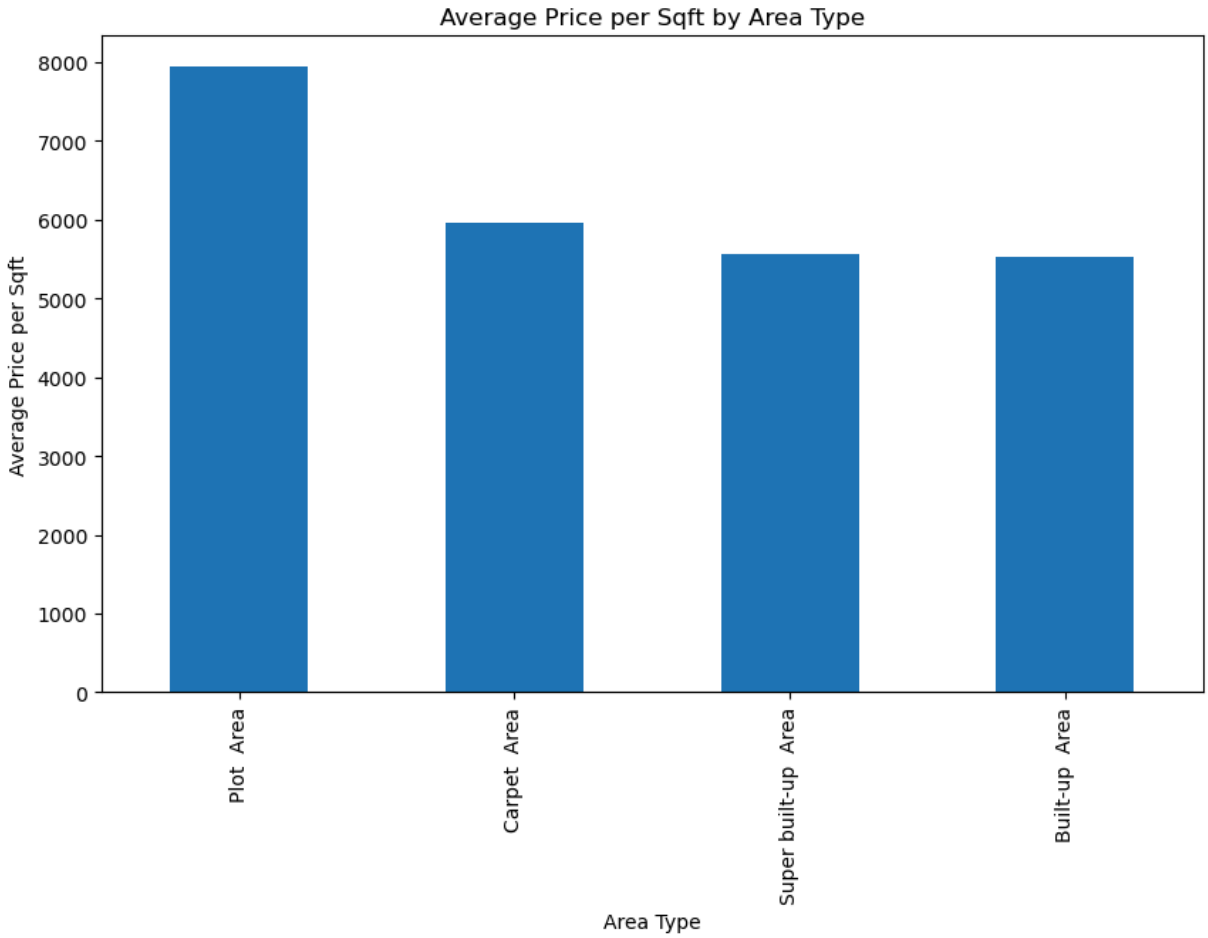
In [56]:
```
plot_scatter_chart(df9, "Hebbal")
```



Bài tập 6*: Thực hiện các câu lệnh để trả lời các câu hỏi dưới đây:

- Thống kê giá nhà theo từng loại khu vực (area_type). Làm với df9:
- xem xét theo từng khu vực, thì giá nhà trung bình (price_per_sqft) là bao nhiêu, tương quan về giá nhà trung bình giữa các khu vực
- Gợi ý: Phần này có thể đưa ra kết quả dạng bảng hoặc biểu đồ (cột, histogram, ...).
- Sử dụng các lệnh: df.groupby(), df.sortvalues(), ... để trích xuất giá trị
- Sử dụng matplotlib: plt.bar(), ...

In [58]:
```
# Thống kê giá nhà trung bình theo area_type
grouped = df9.groupby('area_type')['price_per_sqft'].mean().sort_values(ascending=F
print("Average price per sqft by area type:")
print(grouped)

# Plot bar chart
grouped.plot(kind='bar', figsize=(10,6))
plt.title('Average Price per Sqft by Area Type')
plt.xlabel('Area Type')
plt.ylabel('Average Price per Sqft')
plt.show()
```

```
Average price per sqft by area type:
area_type
Plot  Area              7936.461079
Carpet  Area            5955.799703
Super built-up  Area    5564.771724
Built-up  Area          5533.374127
Name: price_per_sqft, dtype: float64
```